

KRUNAL R. PANCHAL

Combustion researcher — flame and gasification experiments, chemical kinetics, kinetic-model development and uncertainty quantification

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RESEARCH PROFILE

My PhD developed MOSIAC, an open-source, multi-objective optimization framework for combustion kinetic-mechanism development — integrating automated mechanism generation, mechanism reduction and reduced-order/surrogate modelling, Bayesian uncertainty quantification, multi-stage kinetics-informed calibration, and machine-learning-accelerated surrogates (§5–6). Alongside this methodological work, I have hands-on combustion-experiment experience: pool-flame characterization of n-heptane as a benchmark fuel (extending to MtG surrogate fuel, with computer-vision-driven video post-processing), a planned counterflow non-premixed burner campaign, and steam/catalyst-assisted downdraft fixed-bed gasification during my M.Tech (§4). I also work at a developing, working level in quantum chemical computation (Gaussian) for reaction-rate calculations — an area I would like to grow further, not one I present as my primary focus. I am looking to bring MOSIAC and this combined experimental/kinetics background to a research group working on combustion kinetics, uncertainty quantification, or scientific machine learning for stiff physical systems.

EDUCATION

PhD, Mechanical Engineering — Awarded 16 Jul 2026

Indian Institute of Technology Madras

Thesis: “Development of a Fast, Robust, and Kinetics-Based Optimization Framework for Combustion Kinetic Models” (viva 12 May 2026). Supervisor: Prof. Krithika Narayanaswamy.

M.Tech, Mechanical Engineering (Thermal Engineering) — 2018

Nirma University, Institute of Technology, Ahmedabad

B.E., Mechanical Engineering — 2016

Government Engineering College, Modasa, Gujarat

COMBUSTION EXPERIMENTS: APPARATUS DESIGN, COMMISSIONING AND OPERATION

4.1 Pool flame studies — n-heptane benchmark characterization

Jointly with a fellow doctoral student, Gagan Verma (PhD scholar, IIT Madras), under the Shell project (PIs: Prof. V. Raghavan and Prof. Krithika Narayanaswamy): pool-flame studies characterizing n-heptane as a benchmark fuel, to be extended to MtG (methanol-to-gasoline) surrogate fuel. Campaign ongoing, work through 30 Sep 2026; a journal paper is the expected outcome. Video post-processing for this campaign uses Flame Processor, a computer-vision tool of mine for automatic burner detection and camera-jerk-compensated frame extraction, replacing a hand-edited MATLAB workflow.

4.2 Counterflow non-premixed burner — planned

Planned next stage of the same Shell project: a counterflow non-premixed burner for liquid fuel flow rates from 1 l/min to 10 ml/min (strain rates 20 s^{-1} to 80 s^{-1}), for MtG/N₂/Air mixtures blended with ethanol. Planned studies: soot loading and species/speciation profiles; extinction strain rates may also be measured.

4.3 Flow-boiling vaporizer rig

Shell project, IIT Madras (PIs: Prof. V. Raghavan, Prof. Krithika Narayanaswamy) — the majority of the current project's work: design and sizing of a direct flow-boiling channel vaporizer (6 mm ID, 8 mm OD; induction heating; two-block isothermal design) for evaporating liquid MtG surrogate fuel and its ethanol blends (0–50%), using the Kandlikar correlation for two-phase pressure drop and critical-heat-flux margin. github.com/KrunalPanchal1995/Vaporizer-Design

4.4 Downdraft fixed-bed gasifier — steam- and catalyst-assisted lignite gasification

M.Tech, Nirma University, Institute of Technology, Ahmedabad. Designed and commissioned a 10 kWth downdraft fixed-bed gasifier; operated steam-gasification campaigns with lignite, varying steam-to-biomass ratio (0.0–0.5), equivalence ratio (0.2–0.35) and MgCO₃ catalyst loading; measured syngas composition (H₂, CO, CO₂, CH₄), cold-gas efficiency and LHV. Published across four journal papers (Energy ×2; Journal of Energy and Environmental Sustainability ×2 — see Refereed Journal Publications below). Winner, ISTE National Award for Best M.Tech Thesis 2018, First Prize.

MULTI-OBJECTIVE OPTIMIZATION FRAMEWORK DEVELOPMENT FOR COMBUSTION KINETIC MODELS

The central methodological contribution of my PhD is MOSIAC, an open-source, multi-objective optimization framework for combustion kinetic-mechanism development that I designed and built end to end — from automated mechanism generation,

through Bayesian uncertainty quantification, to multi-stage, surrogate-accelerated calibration against experimental targets. Its components:

5.1 Kinetic-model development, optimization and uncertainty quantification (MOSIAC core)

- MSO: kinetics-informed HTC/LTC decomposition — ~45% objective reduction, 4x lower cost vs. single-stage optimization.
- MUQ: Bayesian UQ for correlated Arrhenius parameters via Cholesky-factorized joint-normal covariance — convex-optimization-based in the continuous case, extended to discrete/constrained optimization for thermo-parameter and reaction selection. Published, Combustion Theory and Modelling, 2024.
- SAC (Sampling of Arrhenius Curves): stratified sampling across three structural classes of rate-constant curves, average MRE 0.2% (ignition delay) and 0.6% (flame speed) across 118/138 combustion targets.
- Correlation-aware partial response surfaces for combustion kinetic-mechanism calibration — manuscript in preparation (target: Fuels).
- A power-mean (Hölder) objective function, annealed during the search, for fair multi-dataset calibration. Accepted, 64th Symposium (Japanese) on Combustion.

5.2 Simulation stack

Cantera, CHEMKIN, FlameMaster, OpenFOAM, ANSYS Fluent/Workbench.

5.3 Automated mechanism generation (MOSIAC mechanism-generation module)

RMG-Py-based generation and pre/post-processing of detailed combustion kinetic mechanisms for H₂, H₂/CO and NH₃ systems, for downstream CFD use.

5.4 Quantum chemistry — working level, developing

Working level: builds molecular structures (Avogadro) and runs geometry-optimization and frequency calculations in Gaussian, obtains partition functions, and derives rate constants for small reaction systems. I understand what these calculations solve for and their role in kinetic-model construction, and this is an area I want to develop further under guidance.

MACHINE LEARNING FOR COMBUSTION REACTION MODELS

Machine learning runs through both my PhD framework and a separate, actively-led research line:

- Surrogate/response-surface and neural models are core, load-bearing components of MOSIAC itself (§5.1) — they are what make its multi-stage optimization tractable at the cost MOSIAC achieves.
- Leading an integrator-free neural surrogate for stiff H₂/O₂ ignition-chemistry kinetics (title TBD). In preparation, target NeurIPS 2027.
- Leading a second, distinct submission combining NeuralODE and neural-operator architectures with reinforcement-learning/heuristic-algorithm search on the same problem (title TBD). In preparation, target NeurIPS 2027 (second, separate submission).
- MOSIAC's mechanism-generation module (§5.3) automates RMG-Py-based mechanism generation and pre/post-processing for downstream kinetic-model use.

REFEREED JOURNAL PUBLICATIONS AND CONTRIBUTION STATEMENTS

Author position given for every entry; contributions described using CRediT taxonomy roles.

Published (5)

Panchal K., Vasudevan V., Ambikasaran S., Narayanaswamy K. "A fast uncertainty quantification methodology and sampling technique for joint probability distribution of the Arrhenius rate expression: a case study applied to H₂/CO kinetic mechanism." Combustion Theory and Modelling 28(4):411–440 (2024). *[Published, refereed]*

Author position: 1 of 4.

DOI: 10.1080/13647830.2024.2326413.

Contribution: Conceptualization; Methodology; Software; Formal analysis; Investigation; Writing – original draft.

Upadhyay D.S., **Panchal K.R.**, Sakhiya A.K.V., Patel R.N. "Air-Steam gasification of lignite in a fixed bed gasifier: Influence of steam to lignite ratio on performance of downdraft gasifier." Energy 211:118187 (2020). *[Published, refereed]*

Author position: 2 of 4.

DOI: 10.1016/j.energy.2020.118187.

Contribution: Investigation — operated the downdraft gasifier for the steam-ratio campaigns and acquired syngas-composition and temperature data; Formal analysis — reduced the performance data; Writing – review & editing.

Upadhyay D.S., Sakhiya A.K., **Panchal K.**, Patel A.H., Patel R.N. "Effect of equivalence ratio on the performance of the downdraft gasifier — An experimental and modelling approach." Energy 168:833–846 (2019). *[Published, refereed]*

Author position: 3 of 5.

DOI: 10.1016/j.energy.2018.11.133.

Contribution: Investigation — operated the gasifier across the equivalence-ratio sweep; Software — contributed to the equilibrium-based performance model; Writing – review & editing.

Sakhiya A.K., Chaturvedi A., Barasara V., **Panchal K.**, Upadhyay D.S., Patel R.N. "Elemental and heat loss analysis of different feedstock in downdraft gasifier." *Journal of Energy and Environmental Sustainability* 7:10–16 (2019). *[Published, refereed]*

Author position: 4 of 6.

DOI: 10.47469/jees.2019.v07.100070.

Contribution: Investigation — contributed gasifier operating data across feedstocks; Data curation.

Upadhyay D.S., Sakhiya A.K.V., **Panchal K.R.**, Patel R.N. "Thermodynamic analysis of lignite gasification in the downdraft gasifier." *Journal of Energy and Environmental Sustainability* 5:58–64 (2018). *[Published, refereed]*

Author position: 3 of 4.

DOI: 10.47469/jees.2018.v05.100058.

Contribution: Investigation — operated the gasifier and acquired thermodynamic performance data; Formal analysis.

Under review (1) — not yet refereed

Singh B., **Panchal K.**, Narayanaswamy K., Ambikasaran S. "A novel, temperature-dependent optimization framework for thermodynamic parameters of fuel kinetic mechanism." Submitted to *Fuel*, 12 Aug 2026. *[Under review — not yet refereed]*

Author position: 2 of 4.

Manuscript ID: JFUE-D-26-10471.

Contribution: Methodology; Software; Formal analysis; Writing – review & editing.

REFEREED CONFERENCE PUBLICATIONS

Accepted (3) and published (1); refereed conference proceedings, listed separately from the journal record above.

Panchal K.R. * (corresponding), Narayanaswamy K. "A Power-Mean (Hölder) Objective Function for Balancing Unequal Datasets in Combustion Kinetic Mechanism Optimization: Case Studies on Methyl Butanoate and n-Heptane." 64th Symposium (Japanese) on Combustion, Fukuoka International Congress Center, Fukuoka, Japan, 4–6 Nov 2026. *[Accepted (acceptance notified 7 Aug 2026)]*

Author position: 1 of 2.

Srikanth G. *, **Panchal K.**, Narayanaswamy K. "A Novel Adaptive Curve-Capturing Technique for Optimizing Fuel Reaction Kinetic Mechanisms Against Shock Tube Species Concentration Profiles Using Polynomial Response Surfaces." 64th Symposium (Japanese) on Combustion, Fukuoka, Japan, 4–6 Nov 2026. *[Accepted (acceptance notified 7 Aug 2026)]*

Author position: 2 of 3.

Panchal K., Vaisakh V., Narayanaswamy K. "A multi stage kinetic reaction optimization strategy applied to n-heptane combustion mechanism." 14th Asia-Pacific Conference on Combustion (ASPACC), 2023. *[Published]*

Author position: 1 of 3.

MANUSCRIPTS IN PREPARATION

- Panchal K., Narayanaswamy K. "Kinetics-Based Multi-Stage Optimization with Correlation-Aware Partial Response Surfaces for Combustion Kinetic Mechanisms: Case Studies Applied to 1,3-Butadiene, Methyl Butanoate, and n-Heptane Models." Target: *Fuels*.
- Panchal K., Narayanaswamy K. "Resolving Dataset Dominance in Combustion Kinetic Mechanism Optimization via Annealed Hölder-Mean Scalarization." Target: *Combustion Theory and Modelling*.
- Panchal K., Singh B., Srikanth G., Vaisakh V., Narayanaswamy K., Ambikasaran S. "MOSIAC: an open-source workbench for uncertainty-quantified optimization of combustion reaction mechanisms." Target: *Journal of Open Source Software (JOSS)*.
- Srikanth G., Panchal K., Narayanaswamy K. "Uncertainty-Quantified, Correlation-Aware Optimization of Combustion Kinetic Mechanisms Against Shock-Tube, Jet-Stirred, and Plug-Flow Reactor Species Concentration Profiles (working title)." Target: *Combustion and Flame*.
- Panchal K., et al. "Joint Uncertainty Quantification and Optimization of Arrhenius Parameters for Combustion Kinetic Models via Correlation-Aware Partial Polynomial Response Surfaces." Target: *Combustion and Flame*.
- Two further manuscripts targeting NeurIPS 2027 on stiff H₂/O₂ ignition-chemistry surrogates: an integrator-free neural surrogate architecture, and a separate NeuralODE + neural-operator + RL/heuristic-search architecture (both led by Krunal Panchal; both titles TBD).

PROFESSIONAL EXPERIENCE

Project Associate — Shell Project

Indian Institute of Technology Madras

Pls: Prof. V. Raghavan, Prof. Krithika Narayanaswamy.

Jun 2026 – Present

Chennai, India

- Pool-flame benchmarking and planned counterflow campaign (§4.1–4.2); flow-boiling vaporizer design, the majority of current project work (§4.3).

Project Associate — AI/ML Chess Analysis

Jul 2025 – Apr 2026

Indian Institute of Technology Madras

Chennai, India

- Full-stack chess-engine analysis platform (non-combustion).

Project Assistant

Aug 2024 – Jun 2025

Indian Institute of Technology Madras

Chennai, India

- Extended MUQ-SAC to thermodynamic-parameter optimization and multi-fuel validation (H₂/CO, propane).

Research Scholar (PhD)

Aug 2018 – Jul 2026

Indian Institute of Technology Madras

Chennai, India

- Designed and built MOSIAC, an open-source multi-objective optimization framework for combustion kinetic-mechanism development (§5–6); pool-flame and planned counterflow experiments (§4.1–4.2).
- Decarbonisation Technology course project (Aug–Dec 2022, team of six; my scope: base-case tuyere dynamics and the design of experiments): ANSYS Fluent reacting-flow study of H₂/CH₄ co-injection with pulverised coal in blast-furnace tuyeres. CO₂ intensity fell from 0.121 to 0.067 gCO₂/kJ at 80% H₂ substitution (0.091 for CH₄); peak temperatures of 3040–4500 K exceeded the 2400 K material limit, showing the substitution ceiling is set by tuyere materials, not chemistry.

Graduate Research Assistant (M.Tech)

Jul 2017 – Jun 2018

Nirma University, Institute of Technology

Ahmedabad, India

- Downdraft fixed-bed gasifier design, commissioning and operation (§4.4).

TECHNICAL SKILLS

Combustion apparatus: pool-flame benchmarking and planned counterflow non-premixed burner work (§4.1–4.2); flow-boiling vaporizer design (§4.3); syngas composition analysis on the downdraft gasifier (§4.4).

Combustion simulation: Cantera, CHEMKIN, FlameMaster, OpenFOAM, ANSYS Fluent/Workbench.

Development of gas-phase chemical kinetic models: MOSIAC — RMG-Py-based mechanism generation and pre/post-processing for H₂, H₂/CO and NH₃ systems, mechanism reduction, reduced-order/surrogate modelling, and experimental/mechanism data pre-processing tools (§5–6).

Quantum chemical computation: Gaussian — working level, developing (§5.4).

Programming: Python (NumPy/SciPy/Matplotlib), MATLAB, C++, Git/GitHub.

OPEN-SOURCE SOFTWARE

MOSIAC — a unified, open-source multi-objective optimization framework for combustion kinetic-mechanism development, integrating:

- Mechanism generation and pre/post-processing — RMG-Py-based, for H₂, H₂/CO and NH₃ systems.
- MUQ-SAC — Bayesian uncertainty quantification (MUQ) and stratified Arrhenius-curve sampling (SAC). github.com/KrunalPanchal1995/MUQ-SAC
- mechanism-optimization-suite — multi-stage kinetics-informed optimization (MSO), power-mean/Hölder objective function, correlation-aware partial-PRS surrogates. github.com/KrunalPanchal1995/mechanism-optimization-suite
- Mechanism-reduction and experimental/mechanism data pre-processing tools.
- MOSIAC — unified workbench integrating all of the above; in preparation for JOSS. github.com/KrunalPanchal1995/MOSIAC
- Vaporizer-Design — sizing tool for the flow-boiling vaporizer rig (§4.3). github.com/KrunalPanchal1995/Vaporizer-Design
- Flame Processor — computer-vision tool for automatic burner detection and camera-jerk-compensated video post-processing on the pool-flame campaign (§4.1).

TEACHING

Teaching Assistant/Lab Demonstrator, IIT Madras: Engineering Drawing (2021), Combustion Technology Lab (2022), Mechanical Lab I (2020).

HONOURS AND AWARDS

- ISTE National Award for Best M.Tech Thesis 2018, First Prize — Indian Society for Technical Education.
- MHRD (Govt. of India) Scholarship — PhD at IIT Madras.

AVAILABILITY

Available to start after 30 November 2026, upon completion of my current appointment.